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MCY 612
Lab 4 Securing EFS Storage with KMS
April, 2025

MCY 612 Cloud Security
Lab#4: Securing EFS Storage with KMS

In this lab, you will study the KMS and EFS. You will use the AWS CLI to create a KMS key and then encrypt an EFS file system with the KMS key to secure the storage.

Tools and Materials

AWS credentials

AWS CloudShell CLI

Amazon Linux 2 AMI: ami-087c17d1fe0178315

What is id of the VPC that you created in lab 2?

What is id of the subnet that you created in lab 2?

What is id of the security group that you created for EC2 instances in lab 2?

Note: You need to replace words in italics with your information.

1. Start your AWS CloudShell terminal. Create a KMS key with the following command:

`aws kms create-key --description "The key is used for EFS encryption"`

What is your command and its complete output? What is your KeyId?

"KeyId": "abb875c0-334e-4cad-8d69-c0b0f456156a",

```
~ $ aws kms create-key --description "The key is used for EFS encryption"
{
  "KeyMetadata": {
    "AWSAccountId": "881490118823",
    "KeyId": "abb875c0-334e-4cad-8d69-c0b0f456156a",
    "Arn": "arn:aws:kms:us-east-1:881490118823:key/abb875c0-334e-4cad-8d69-c0b0f456156a",
    "CreationDate": "2025-04-13T00:19:52.965000+00:00",
    "Enabled": true,
    "Description": "The key is used for EFS encryption",
    "KeyUsage": "ENCRYPT_DECRYPT",
    "KeyState": "Enabled",
    "Origin": "AWS_KMS",
    "KeyManager": "CUSTOMER",
    "CustomerMasterKeySpec": "SYMMETRIC_DEFAULT",
    "KeySpec": "SYMMETRIC_DEFAULT",
    ... skipping ...
  }
  "KeyMetadata": {
    "AWSAccountId": "881490118823",
    "KeyId": "abb875c0-334e-4cad-8d69-c0b0f456156a",
    "Arn": "arn:aws:kms:us-east-1:881490118823:key/abb875c0-334e-4cad-8d69-c0b0f456156a",
    "CreationDate": "2025-04-13T00:19:52.965000+00:00",
    "Enabled": true,
    "Description": "The key is used for EFS encryption",
    "KeyUsage": "ENCRYPT_DECRYPT",
    "KeyState": "Enabled",
    "Origin": "AWS_KMS",
    "KeyManager": "CUSTOMER",
    "CustomerMasterKeySpec": "SYMMETRIC_DEFAULT",
```

```

    "KeySpec": "SYMMETRIC_DEFAULT",
    ...skipping...
  {
    "KeyMetadata": {
      "AWSAccountId": "881490118823",
      "KeyId": "abb875c0-334e-4cad-8d69-c0b0f456156a",
      "Arn": "arn:aws:kms:us-east-1:881490118823:key/abb875c0-334e-4cad-8d69-c0b0f456156a",
      "CreationDate": "2025-04-13T00:19:52.965000+00:00",
      "Enabled": true,
      "Description": "The key is used for EFS encryption",
      "KeyUsage": "ENCRYPT_DECRYPT",
      "KeyState": "Enabled",
      "Origin": "AWS_KMS",
      "KeyManager": "CUSTOMER",
      "CustomerMasterKeySpec": "SYMMETRIC_DEFAULT",
      "KeySpec": "SYMMETRIC_DEFAULT",
      "EncryptionAlgorithms": [
        "SYMMETRIC_DEFAULT"
      ]
    }
    ...skipping...
  {
    "KeyMetadata": {
      "AWSAccountId": "881490118823",
      "KeyId": "abb875c0-334e-4cad-8d69-c0b0f456156a",
      "Arn": "arn:aws:kms:us-east-1:881490118823:key/abb875c0-334e-4cad-8d69-c0b0f456156a",
      "CreationDate": "2025-04-13T00:19:52.965000+00:00",
      "Enabled": true,
      "Description": "The key is used for EFS encryption",
      "KeyUsage": "ENCRYPT_DECRYPT",
      "KeyState": "Enabled",

```

CloudShell Feedback

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```

      "Origin": "AWS_KMS",
      "KeyManager": "CUSTOMER",
      "CustomerMasterKeySpec": "SYMMETRIC_DEFAULT",
      "KeySpec": "SYMMETRIC_DEFAULT",
      "EncryptionAlgorithms": [
        "SYMMETRIC_DEFAULT"
      ]
    }
    ...skipping...
    ...skipping...
  {
    "KeyMetadata": {
      "AWSAccountId": "881490118823",
      "KeyId": "abb875c0-334e-4cad-8d69-c0b0f456156a",
      "Arn": "arn:aws:kms:us-east-1:881490118823:key/abb875c0-334e-4cad-8d69-c0b0f456156a",
      "CreationDate": "2025-04-13T00:19:52.965000+00:00",
      "Enabled": true,
      "Description": "The key is used for EFS encryption",
      "KeyUsage": "ENCRYPT_DECRYPT",
      "KeyState": "Enabled",
      "Origin": "AWS_KMS",
      "KeyManager": "CUSTOMER",
      "CustomerMasterKeySpec": "SYMMETRIC_DEFAULT",
      "KeySpec": "SYMMETRIC_DEFAULT",
      "EncryptionAlgorithms": [
        "SYMMETRIC_DEFAULT"
      ]
    }
    ...skipping...
  {
    "KeyMetadata": {

```

```

    "AWSAccountId": "881490118823",
    "KeyId": "abb875c0-334e-4cad-8d69-c0b0f456156a",
    "Arn": "arn:aws:kms:us-east-1:881490118823:key/abb875c0-334e-4cad-8d69-c0b0f456156a",
    "CreationDate": "2025-04-13T00:19:52.965000+00:00",
    "Enabled": true,
    "Description": "The key is used for EFS encryption",
    "KeyUsage": "ENCRYPT_DECRYPT",
    "KeyState": "Enabled",
    "Origin": "AWS_KMS",
    "KeyManager": "CUSTOMER",
    "CustomerMasterKeySpec": "SYMMETRIC_DEFAULT",
    "KeySpec": "SYMMETRIC_DEFAULT",
    "EncryptionAlgorithms": [
        "SYMMETRIC_DEFAULT"
    ],
    "MultiRegion": false
  },
  ...skipping...
  {
    "KeyMetadata": {
      "AWSAccountId": "881490118823",
      "KeyId": "abb875c0-334e-4cad-8d69-c0b0f456156a",
      "Arn": "arn:aws:kms:us-east-1:881490118823:key/abb875c0-334e-4cad-8d69-c0b0f456156a",
      "CreationDate": "2025-04-13T00:19:52.965000+00:00",
      "Enabled": true,
      "Description": "The key is used for EFS encryption",
      "KeyUsage": "ENCRYPT_DECRYPT",
      "KeyState": "Enabled",
      "Origin": "AWS_KMS",

```

AWS CloudShell

```

      "KeyManager": "CUSTOMER",
      "CustomerMasterKeySpec": "SYMMETRIC_DEFAULT",
      "KeySpec": "SYMMETRIC_DEFAULT",
      "EncryptionAlgorithms": [
        "SYMMETRIC_DEFAULT"
      ],
      "MultiRegion": false
    }
  },
  ...skipping...
  ...skipping...
  {
    "KeyMetadata": {
      "AWSAccountId": "881490118823",
      "KeyId": "abb875c0-334e-4cad-8d69-c0b0f456156a",
      "Arn": "arn:aws:kms:us-east-1:881490118823:key/abb875c0-334e-4cad-8d69-c0b0f456156a",
      "CreationDate": "2025-04-13T00:19:52.965000+00:00",
      "Enabled": true,
      "Description": "The key is used for EFS encryption",
      "KeyUsage": "ENCRYPT_DECRYPT",
      "KeyState": "Enabled",
      "Origin": "AWS_KMS",
      "KeyManager": "CUSTOMER",
      "CustomerMasterKeySpec": "SYMMETRIC_DEFAULT",
      "KeySpec": "SYMMETRIC_DEFAULT",
      "EncryptionAlgorithms": [
        "SYMMETRIC_DEFAULT"
      ],
      "MultiRegion": false
    }
  },
  ...skipping...

```

AWS CloudShell

```

    }
  }
  (END)...skipping...
  {
    "KeyMetadata": {
      "AWSAccountId": "881490118823",
      "KeyId": "abb875c0-334e-4cad-8d69-c0b0f456156a",
      "Arn": "arn:aws:kms:us-east-1:881490118823:key/abb875c0-334e-4cad-8d69-c0b0f456156a",
      "CreationDate": "2025-04-13T00:19:52.965000+00:00",
      "Enabled": true,
      "Description": "The key is used for EFS encryption",
      "KeyUsage": "ENCRYPT_DECRYPT",
      "KeyState": "Enabled",
      "Origin": "AWS_KMS",
      "KeyManager": "CUSTOMER",
      "CustomerMasterKeySpec": "SYMMETRIC_DEFAULT",
      "KeySpec": "SYMMETRIC_DEFAULT",
      "EncryptionAlgorithms": [
        "SYMMETRIC_DEFAULT"
      ],
      "MultiRegion": false
    }
  }
}
}
~
~
~
(END)...skipping...
{
  "KeyMetadata": {
    "AWSAccountId": "881490118823",

```

```

"KeyId": "abb875c0-334e-4cad-8d69-c0b0f456156a",
"Arn": "arn:aws:kms:us-east-1:881490118823:key/abb875c0-334e-4cad-8d69-c0b0f456156a",
"CreationDate": "2025-04-13T00:19:52.965000+00:00",
"Enabled": true,
"Description": "The key is used for EFS encryption",
"KeyUsage": "ENCRYPT_DECRYPT",
"KeyState": "Enabled",
"Origin": "AWS_KMS",
"KeyManager": "CUSTOMER",
"CustomerMasterKeySpec": "SYMMETRIC_DEFAULT",
"KeySpec": "SYMMETRIC_DEFAULT",
"EncryptionAlgorithms": [
  "SYMMETRIC_DEFAULT"
],
"MultiRegion": false
}
}
~
~
~
(END)...skipping...
(END)...skipping...
{
  "KeyMetadata": {
    "AWSAccountId": "881490118823",
    "KeyId": "abb875c0-334e-4cad-8d69-c0b0f456156a",
    "Arn": "arn:aws:kms:us-east-1:881490118823:key/abb875c0-334e-4cad-8d69-c0b0f456156a",
    "CreationDate": "2025-04-13T00:19:52.965000+00:00",
    "Enabled": true,

```



```

    "KeySpec": "SYMMETRIC_DEFAULT",
    "EncryptionAlgorithms": [
        "SYMMETRIC_DEFAULT"
    ],
    "MultiRegion": false
}
~
~
~
~
~
~
~ $ █

```

2. Create an alias, a friendly name, for the KMS key. The alias will be used to encrypt the EFS filesystem. Run the following command to create the alias:

`aws kms create-alias --alias-name alias/carrd12 --target-key-id your_KeyId`

What is your command?

```

~ $ aws kms create-alias --alias-name alias/carrd12 --target-key-id abb875c0-334e-4cad-8d69-c0b0f456156a
~ $ █

```

3. Get detailed information about the KMS key you create via the following command:

`aws kms describe-key --key-id alias/carrd12
abb875c0-334e-4cad-8d69-c0b0f456156a`

What is your command and its complete output?

```

~ $ aws kms describe-key --key-id alias/carrd12
{
  "KeyMetadata": {
    "AWSAccountId": "881490118823",
    "KeyId": "abb875c0-334e-4cad-8d69-c0b0f456156a",
    "Arn": "arn:aws:kms:us-east-1:881490118823:key/abb875c0-334e-4cad-8d69-c0b0f456156a",
    "CreationDate": "2025-04-13T00:19:52.965000+00:00",
    "Enabled": true,
    "Description": "The key is used for EFS encryption",
    "KeyUsage": "ENCRYPT_DECRYPT",
    "KeyState": "Enabled",
    "Origin": "AWS_KMS",
    "KeyManager": "CUSTOMER",
    "CustomerMasterKeySpec": "SYMMETRIC_DEFAULT",
    "KeySpec": "SYMMETRIC_DEFAULT",
    "EncryptionAlgorithms": [
      "SYMMETRIC_DEFAULT"
    ],
    "MultiRegion": false
  }
}
~ $ █

```

4. Create a new encrypted EFS file system with your KMS key via the following command:

`aws efs create-file-system --encrypted --region us-east-1 --performance-mode generalPurpose --kms-key-id
alias/carrd12`

What is your command and its complete output? What is your FileSystemId?

`"FileSystemId": "fs-049b01ea4cdad1d73",`

```

~ $ aws efs create-file-system --encrypted --region us-east-1 --performance-mode generalPurpose --kms-key-id alias/carrd12
{
  "OwnerId": "881490118823",
  "CreationToken": "7ca387e0-06f5-414b-b369-447f4efa445b",
  "FileSystemId": "fs-049b01ea4cdad1d73",
  "FileSystemArn": "arn:aws:elasticfilesystem:us-east-1:881490118823:file-system/fs-049b01ea4cdad1d73",
  "CreationTime": "2025-04-13T00:36:53+00:00",
  "LifecycleState": "creating",
  "NumberOfMountTargets": 0,
  "SizeInBytes": {
    "Value": 0,
    "ValueInIA": 0,
    "ValueInStandard": 0,
    "ValueInArchive": 0
  },
  "PerformanceMode": "generalPurpose",
  "Encrypted": true,
  "KmsKeyId": "arn:aws:kms:us-east-1:881490118823:key/abb875c0-334e-4cad-8d69-c0b0f456156a",
  "ThroughputMode": "bursting",
  "Tags": [],
  "FileSystemProtection": {
    "ReplicationOverwriteProtection": "ENABLED"
  }
}

```

5. Create a security group for an EFS mount target. The EFS mount target provides an IP address for an NFSv4 endpoint at which you can mount an Amazon EFS file system. Run the following command:

```
aws ec2 create-security-group --region us-east-1 --group-name efs-mount-target-carrd12 --vpc-id your_vpc_id --description "lab 4 EFS Mount Target Security Group"
```

What is your command and its complete output? What is the Id of your security group for the EFS mount target?

Sg-0005fa9262ffcd826

```

~ $ aws ec2 create-security-group --region us-east-1 --group-name efs-mount-target-carrd12 --vpc-id vpc-0cb3e30db617f6175 --description "lab 4 EFS Mount Target Security Group"
{
  "GroupId": "sg-0005fa9262ffcd826",
  "SecurityGroupArn": "arn:aws:ec2:us-east-1:881490118823:security-group/sg-0005fa9262ffcd826"
}

```

6. Add a rule for NFS (port 2049) on the security group for the mount target via the following commands:

```
aws ec2 authorize-security-group-ingress --group-id id_of_your_security_group_for_the_EFS_mount_target --protocol tcp --port 2049 --source-group id_of_your_security_group_for_the_instance --region us-east-1
```

What is your command and its complete output?

```

~ $ aws ec2 authorize-security-group-ingress --group-id sg-0005fa9262ffcd826 --protocol tcp --port 2049 --source-group sg-0b1cfa43fd7ea602b --region us-east-1
{
  "Return": true,
  "SecurityGroupRules": [
    {
      "SecurityGroupRuleId": "sgr-0f4e41e11460a68c4",
      "GroupId": "sg-0005fa9262ffcd826",
      "GroupOwnerId": "881490118823",
      "IsEgress": false,
      "IpProtocol": "tcp",
      "FromPort": 2049,
      "ToPort": 2049,
      "ReferencedGroupInfo": {
        "GroupId": "sg-0b1cfa43fd7ea602b",
        "UserId": "881490118823"
      },
      "SecurityGroupRuleArn": "arn:aws:ec2:us-east-1:881490118823:security-group-rule/sgr-0f4e41e11460a68c4"
    }
  ]
}

```

NOTE: The id_of_your_security_group_for_the_instance is the ID of the security group you created in Lab 2 and listed at the beginning of this lab.

7. Create a mount target for the file system you created at step 4 with the following command:

```
aws efs create-mount-target --file-system-id your_FileSystemId --subnet-id your_subnet-id --security-group id_of_your_security_group_for_the_EFS_mount_target --region us-east-1
```

What is your command and its complete output? What is your MountTargetId?

"MountTargetId": "fsmt-0e176469c39c501f0",

```

~ $ aws efs create-mount-target --file-system-id fs-049b01ea4cdad1d73 --subnet-id subnet-09df9a6645aea6ede --security-group sg-0005fa9262ffcd826 --region us-east-1
{
  "OwnerId": "881490118823",
  "MountTargetId": "fsmt-0e176469c39c501f0",
  "FileSystemId": "fs-049b01ea4cdad1d73",
  "SubnetId": "subnet-09df9a6645aea6ede",
  "LifeCycleState": "creating",
  "IpAddress": "10.6.8.13",
  "NetworkInterfaceId": "eni-0eebb1007773b91c3",
  "AvailabilityZoneId": "us-east-2",
  "AvailabilityZoneName": "us-east-1c",
  "VpcId": "vpc-0cb3e30db617f6175"
}
~ $

```

8. Modify the `enableDnsHostnames` attribute of your VPC. This attribute indicates whether an instance launched in the VPC gets DNS hostnames. You set this attribute to `true` so that the instance in your VPC gets DNS hostnames. The instance needs to have a DNS hostname so that you can mount the EFS file system to the instance. Run the following command:

```
aws ec2 modify-vpc-attribute --vpc-id your_vpc_id --enable-dns-hostnames "{\"Value\":true}"
```

What is your command?

```

~ $ aws ec2 modify-vpc-attribute --vpc-id vpc-0cb3e30db617f6175 --enable-dns-hostnames "{\"Value\":true}"
~ $

```

9. You will use an Amazon Linux EC2 instance for this lab. You need the Amazon EFS client to use the EFS. The Amazon EFS client is installed as part of the `amazon-efs-utils` package, an open-source collection of Amazon EFS tools. The `amazon-efs-utils` package is available in the Amazon Linux package repositories, and you can build and install the package on other Linux distributions. It's easier to install the `amazon-efs-utils` package on Amazon Linux. Create an Amazon Linux EC2 instance in the subnet of your VPC via the following command:

```
aws ec2 run-instances --image-id ami-087c17d1fe0178315 --count 1 --instance-type t3.micro --key-name carrrd12-key --security-group-ids your_securitygroup_id --subnet-id your_subnet_id
```

What command do you use to find out your instance's public IP address? What is your instance's public address?

```

~ $ aws ec2 describe-instances --instance-ids i-006073396eff19540 --query "Reservations[*].Instances[*].PublicIpAddress" --output text
44.204.235.216
~ $

```

```

~ $ aws ec2 run-instances --image-id ami-087c17d1fe0178315 --count 1 --instance-type t3.micro --key-name carrrd12-key --security-group-ids sg-0b1cfa43fd7ea602b --subnet-id subnet-09df9a6645aea6ede
{
  "ReservationId": "r-05d7e8346a131c0b2",
  "OwnerId": "881490118823",
  "Groups": [],
  "Instances": [
    {
      "Architecture": "x86_64",
      "BlockDeviceMappings": [],
      "ClientToken": "ba1de5db-fb9d-4777-9b61-936e005b41ea",
      "EbsOptimized": false,
      "...skipping..."
    }
  ]
}
~ $

```



```
"Instances": [  
  {  
    "Architecture": "x86_64",  
    "BlockDeviceMappings": [],  
    "ClientToken": "ba1de5db-fb9d-4777-9b61-936e005b41ea",  
    "EbsOptimized": false,  
    "EnaSupport": true,  
    "Hypervisor": "xen",  
    "NetworkInterfaces": [  
      {  
        "Attachment": {  
          "AttachTime": "2025-04-13T01:12:49+00:00",  
:...skipping...  
      {  
        "ReservationId": "r-05d7e8346a131c0b2",  
        "OwnerId": "881490118823",  
        "Groups": [],  
        "Instances": [  
          {  
            "Architecture": "x86_64",  
            "BlockDeviceMappings": [],  
            "ClientToken": "ba1de5db-fb9d-4777-9b61-936e005b41ea",  
            "EbsOptimized": false,  
            "EnaSupport": true,  
            "Hypervisor": "xen",  
            "NetworkInterfaces": [  
              {  
                "Attachment": {  
                  "AttachTime": "2025-04-13T01:12:49+00:00",
```

```
        "AttachmentId": "eni-attach-0792dc59eed2ee321",
        "DeleteOnTermination": true,
        "DeviceIndex": 0,
...skipping...
...skipping...
{
  "ReservationId": "r-05d7e8346a131c0b2",
  "OwnerId": "881490118823",
  "Groups": [],
  "Instances": [
    {
      "Architecture": "x86_64",
      "BlockDeviceMappings": [],
      "ClientToken": "ba1de5db-fb9d-4777-9b61-936e005b41ea",
      "EbsOptimized": false,
      "EnaSupport": true,
      "Hypervisor": "xen",
      "NetworkInterfaces": [
        {
          "Attachment": {
            "AttachTime": "2025-04-13T01:12:49+00:00",
            "AttachmentId": "eni-attach-0792dc59eed2ee321",
            "DeleteOnTermination": true,
            "DeviceIndex": 0,
            "Status": "attaching",
...skipping...
{
  "ReservationId": "r-05d7e8346a131c0b2",
  "OwnerId": "881490118823",
```

```
"Groups": [],
"Instances": [
  {
    "Architecture": "x86_64",
    "BlockDeviceMappings": [],
    "ClientToken": "ba1de5db-fb9d-4777-9b61-936e005b41ea",
    "EbsOptimized": false,
    "EnaSupport": true,
    "Hypervisor": "xen",
    "NetworkInterfaces": [
      {
        "Attachment": {
          "AttachTime": "2025-04-13T01:12:49+00:00",
          "AttachmentId": "eni-attach-0792dc59eed2ee321",
          "DeleteOnTermination": true,
          "DeviceIndex": 0,
          "Status": "attaching",
          "NetworkCardIndex": 0
        }
      }
    ]
  }
]
...skipping...
{
  "ReservationId": "r-05d7e8346a131c0b2",
  "OwnerId": "881490118823",
  "Groups": [],
  "Instances": [
    {
      "Architecture": "x86_64",
      "BlockDeviceMappings": [],
      "ClientToken": "ba1de5db-fb9d-4777-9b61-936e005b41ea",
      "EbsOptimized": false,
```

```
    "EbsOptimized": false,
    "EnaSupport": true,
    "Hypervisor": "xen",
    "NetworkInterfaces": [
      {
        "Attachment": {
          "AttachTime": "2025-04-13T01:12:49+00:00",
          "AttachmentId": "eni-attach-0792dc59eed2ee321",
          "DeleteOnTermination": true,
          "DeviceIndex": 0,
          "Status": "attaching",
          "NetworkCardIndex": 0
        },
        ...skipping...
      }
    ],
    "ReservationId": "r-05d7e8346a131c0b2",
    "OwnerId": "881490118823",
    "Groups": [],
    "Instances": [
      {
        "Architecture": "x86_64",
        "BlockDeviceMappings": [],
        "ClientToken": "ba1de5db-fb9d-4777-9b61-936e005b41ea",
        "EbsOptimized": false,
        "EnaSupport": true,
        "Hypervisor": "xen",
        "NetworkInterfaces": [
          {
            "Attachment": {
```

```
        "AttachTime": "2025-04-13T01:12:49+00:00",
        "AttachmentId": "eni-attach-0792dc59eed2ee321",
        "DeleteOnTermination": true,
        "DeviceIndex": 0,
        "Status": "attaching",
        "NetworkCardIndex": 0
    },
    "Description": "",
    "Groups": [
: ...skipping...
{
    "ReservationId": "r-05d7e8346a131c0b2",
    "OwnerId": "881490118823",
    "Groups": [],
    "Instances": [
        {
            "Architecture": "x86_64",
            "BlockDeviceMappings": [],
            "ClientToken": "ba1de5db-fb9d-4777-9b61-936e005b41ea",
            "EbsOptimized": false,
            "EnaSupport": true,
            "Hypervisor": "xen",
            "NetworkInterfaces": [
                {
                    "Attachment": {
                        "AttachTime": "2025-04-13T01:12:49+00:00",
                        "AttachmentId": "eni-attach-0792dc59eed2ee321",
                        "DeleteOnTermination": true,
                        "DeviceIndex": 0,
```

```

        "Status": "attaching",
        "NetworkCardIndex": 0
    },
    "Description": "",
    "Groups": [
        {
: ...skipping...
{
    "ReservationId": "r-05d7e8346a131c0b2",
    "OwnerId": "881490118823",
    "Groups": [],
    "Instances": [
        {
            "Architecture": "x86_64",
            "BlockDeviceMappings": [],
            "ClientToken": "ba1de5db-fb9d-4777-9b61-936e005b41ea",
            "EbsOptimized": false,
            "EnaSupport": true,
            "Hypervisor": "xen",
            "NetworkInterfaces": [
                {
                    "Attachment": {
                        "AttachTime": "2025-04-13T01:12:49+00:00",
                        "AttachmentId": "eni-attach-0792dc59eed2ee321",
                        "DeleteOnTermination": true,
                        "DeviceIndex": 0,
                        "Status": "attaching",
                        "NetworkCardIndex": 0
                    },

```

```
        "Description": "",
        "Groups": [
            {
                "GroupId": "sg-0b1cfa43fd7ea602b",
: ...skipping...
{
    "ReservationId": "r-05d7e8346a131c0b2",
    "OwnerId": "881490118823",
    "Groups": [],
    "Instances": [
        {
            "Architecture": "x86_64",
            "BlockDeviceMappings": [],
            "ClientToken": "ba1de5db-fb9d-4777-9b61-936e005b41ea",
            "EbsOptimized": false,
            "EnaSupport": true,
            "Hypervisor": "xen",
            "NetworkInterfaces": [
                {
                    "Attachment": {
                        "AttachTime": "2025-04-13T01:12:49+00:00",
                        "AttachmentId": "eni-attach-0792dc59eed2ee321",
                        "DeleteOnTermination": true,
                        "DeviceIndex": 0,
                        "Status": "attaching",
                        "NetworkCardIndex": 0
                    },
                    "Description": "",
                    "Groups": [
```

AWS CloudShell

```
        {
          "GroupId": "sg-0b1cfa43fd7ea602b",
          "GroupName": "carrd12"
        }
      ],
      "Ipv6Addresses": [],
      "MacAddress": "12:a5:10:4f:91:43",
      "NetworkInterfaceId": "eni-047da17f507ca3fa0",
      "OwnerId": "881490118823",
      "PrivateDnsName": "ip-10-6-8-6.ec2.internal",
      "PrivateIpAddress": "10.6.8.6",
      "PrivateIpAddresses": [
        {
          "Primary": true,
          "PrivateDnsName": "ip-10-6-8-6.ec2.internal",
          "PrivateIpAddress": "10.6.8.6"
        }
      ],
      "SourceDestCheck": true,
      "Status": "in-use",
      "SubnetId": "subnet-09df9a6645aea6ede",
      "VpcId": "vpc-0cb3e30db617f6175",
      "InterfaceType": "interface",
      "Operator": {
        "Managed": false
      }
    }
  ],
  "RootDeviceName": "/dev/xvda",
```



```
"RootDeviceType": "ebs",
"SecurityGroups": [
  {
    "GroupId": "sg-0b1cfa43fd7ea602b",
    "GroupName": "carrd12"
  }
],
"SourceDestCheck": true,
"StateReason": {
  "Code": "pending",
  "Message": "pending"
},
"VirtualizationType": "hvm",
"CpuOptions": {
  "CoreCount": 1,
  "ThreadsPerCore": 2
},
"CapacityReservationSpecification": {
  "CapacityReservationPreference": "open"
},
"MetadataOptions": {
  "State": "pending",
  "HttpTokens": "optional",
  "HttpPutResponseHopLimit": 1,
  "HttpEndpoint": "enabled",
  "HttpProtocolIpv6": "disabled",
  "InstanceMetadataTags": "disabled"
},
"EnclaveOptions": {
```

```
    "Enabled": false
  },
  "PrivateDnsNameOptions": {
    "HostnameType": "ip-name",
    "EnableResourceNameDnsARecord": false,
    "EnableResourceNameDnsAAAARecord": false
  },
  "MaintenanceOptions": {
    "AutoRecovery": "default"
  },
  "CurrentInstanceBootMode": "legacy-bios",
  "Operator": {
    "Managed": false
  },
  "InstanceId": "i-006073396eff19540",
  "ImageId": "ami-087c17d1fe0178315",
  "State": {
    "Code": 0,
    "Name": "pending"
  },
  "PrivateDnsName": "ip-10-6-8-6.ec2.internal",
  "PublicDnsName": "",
  "StateTransitionReason": "",
  "KeyName": "carrd12-key",
  "AmiLaunchIndex": 0,
  "ProductCodes": [],
  "InstanceType": "t3.micro",
  "LaunchTime": "2025-04-13T01:12:49+00:00",
  "Placement": {
```

```

    "GroupName": "",
    "Tenancy": "default",
    "AvailabilityZone": "us-east-1c"
  },
  "Monitoring": {
    "State": "disabled"
  },
  "SubnetId": "subnet-09df9a6645aea6ede",
  "VpcId": "vpc-0cb3e30db617f6175",
  "PrivateIpAddress": "10.6.8.6"
}
]
}
~ $

```

10. Run an SSH command to log into the Amazon Linux instance (ec2-user is the default username) via the following command:

```
ssh -i devenv-key.pem ec2-user@the public IP address of your instance
```

What is your SSH command and its complete output?

```

~ $ ssh -i devenv-key.pem ec2-user@44.204.235.216
The authenticity of host '44.204.235.216 (44.204.235.216)' can't be established.
ED25519 key fingerprint is SHA256:ifa9YguflfwbfDQJD9603jM80Qnu6PhkxAKt8wmdVc8.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '44.204.235.216' (ED25519) to the list of known hosts.

  _ _ | _ _ | _ _ )
 _ | ( / Amazon Linux 2 AMI
 _ _ | \ _ _ | _ _ |

https://aws.amazon.com/amazon-linux-2/
55 package(s) needed for security, out of 105 available
Run "sudo yum update" to apply all updates.
[ec2-user@ip-10-6-8-6 ~]$

```

11. After log in, run the following commands to install the amazon-efs-utils package for the instance.

```
sudo -s
yum install -y amazon-efs-utils
```

12. Create a mount point and then mount the EFS file system on your EC2 instances via the following commands:

```
mkdir /var/efs-encrypted
mount -t efs your_FileSystemId:/var/efs-encrypted
```

What are your commands?

```

[root@ip-10-6-8-6 ec2-user]# mount -t efs fs-049b01ea4cdad1d73:/ /var/efs-encrypted
[root@ip-10-6-8-6 ec2-user]#

```

13. Run the following command:

```
mount
```

What is the complete output of the mount command?

```

[root@ip-10-6-8-6 ec2-user]# mount
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,relatime)
proc on /proc type proc (rw,nosuid,nodev,noexec,relatime)
devtmpfs on /dev type devtmpfs (rw,nosuid,size=479200k,nr_inodes=119800,mode=755)
securityfs on /sys/kernel/security type securityfs (rw,nosuid,nodev,noexec,relatime)
tmpfs on /dev/shm type tmpfs (rw,nosuid,nodev)
devpts on /dev/pts type devpts (rw,nosuid,noexec,relatime,gid=5,mode=620,ptmxmode=000)
tmpfs on /run type tmpfs (rw,nosuid,nodev,mode=755)
tmpfs on /sys/fs/cgroup type tmpfs (ro,nosuid,nodev,noexec,mode=755)
cgroup on /sys/fs/cgroup/systemd type cgroup (rw,nosuid,nodev,noexec,relatime,xattr,release_agent=/usr/lib/systemd/systemd-cgroups-agent,name=systemd)
pstore on /sys/fs/pstore type pstore (rw,nosuid,nodev,noexec,relatime)
cgroup on /sys/fs/cgroup/cpu,cpuacct type cgroup (rw,nosuid,nodev,noexec,relatime,cpu,cpuacct)
cgroup on /sys/fs/cgroup/net_cls,net_prio type cgroup (rw,nosuid,nodev,noexec,relatime,net_cls,net_prio)
cgroup on /sys/fs/cgroup/freezer type cgroup (rw,nosuid,nodev,noexec,relatime,freezer)
cgroup on /sys/fs/cgroup/blkio type cgroup (rw,nosuid,nodev,noexec,relatime,blkio)
cgroup on /sys/fs/cgroup/memory type cgroup (rw,nosuid,nodev,noexec,relatime,memory)
cgroup on /sys/fs/cgroup/cpuset type cgroup (rw,nosuid,nodev,noexec,relatime,cpuset)
cgroup on /sys/fs/cgroup/pids type cgroup (rw,nosuid,nodev,noexec,relatime,pids)
cgroup on /sys/fs/cgroup/hugetlb type cgroup (rw,nosuid,nodev,noexec,relatime,hugetlb)
cgroup on /sys/fs/cgroup/perf_event type cgroup (rw,nosuid,nodev,noexec,relatime,perf_event)
cgroup on /sys/fs/cgroup/devices type cgroup (rw,nosuid,nodev,noexec,relatime,devices)
/dev/mme0n1p1 on / type xfs (rw,nodata,attr2,inode64,noquota)
mqueue on /dev/mqueue type mqueue (rw,relatime)
hugetlbfs on /dev/hugepages type hugetlbfs (rw,relatime,pagesize=2M)
debugfs on /sys/kernel/debug type debugfs (rw,relatime)
sunrpc on /var/lib/nfs/rpc_pipefs type rpc_pipefs (rw,relatime)
systemd-1 on /proc/sys/fs/binfmt_misc type autofs (rw,relatime,fd=27,pgrp=1,timeout=0,minproto=5,maxproto=5,direct,pipe_ino=25371)
tmpfs on /run/user/1000 type tmpfs (rw,nosuid,nodev,relatime,size=97996k,mode=700,uid=1000,gid=1000)
127.0.0.1:/ on /var/efs-encrypted type nfs4 (rw,relatime,vers=4.1,rsize=1048576,wsize=1048576,namlen=255,hard,noreportport,proto=tcp,port=20672,timeo=600,retran
s=2,sec=sys,clientaddr=127.0.0.1,local_lock=none,addr=127.0.0.1)
[root@ip-10-6-8-6 ec2-user]#

```

Part 3: Cleanup

14. Unmount the EFS file system on the EC2 instance via the following command:

```
umount /var/efs-encrypted
```

15. To terminate the instance, you need to exit out of the EC2 instance by typing `exit`, `enter`, `exit`, `enter`. Terminate your instance on the CloudShell terminal.

What is your command and its complete output?

```

[root@ip-10-6-8-6 ec2-user]# exit
exit
[ec2-user@ip-10-6-8-6 ~]$ exit
logout
Connection to 44.204.235.216 closed.
~ $

```

16. Delete your mount target via the following command:

```
aws efs delete-mount-target --mount-target-id your_mount_target_id
```

What is your command?

```

~ $ aws efs delete-mount-target --mount-target-id fsmt-0e176469c39c501f0
~ $

```

17. Delete your EFS file system via the following command:

```
aws efs delete-file-system --file-system-id your_FileSystemId
```

What is your command?

```

~ $ aws efs delete-file-system --file-system-id fs-049b01ea4cdad1d73
~ $

```

18. Delete your KMS alias via the following command:

```
aws kms delete-alias --alias-name alias/your_username
```

What is your command?

```
~ $ aws kms delete-alias --alias-name alias/carrd12
```

19. Schedule the deletion of a KMS key. By default, KMS applies a waiting period of 30 days, but you can specify a waiting period of 7-30 days.

```
aws kms schedule-key-deletion --key-id your_KeyId --pending-window-in-days 7
```

What is your command and its complete output?

```
~ $ aws efs delete-mount-target --mount-target-id fsmt-0e176469c39c501f0
~ $ aws efs delete-file-system --file-system-id fs-049b01ea4cdad1d73
~ $ aws kms delete-alias --alias-name alias/carrd12
~ $ aws kms schedule-key-deletion --key-id abb875c0-334e-4cad-8d69-c0b0f456156a --pending-window-in-days 7
{
  "KeyId": "arn:aws:kms:us-east-1:881490118823:key/abb875c0-334e-4cad-8d69-c0b0f456156a",
  "DeletionDate": "2025-04-20T02:06:46.280000+00:00",
  "KeyState": "PendingDeletion",
  "PendingWindowInDays": 7
}
~ $
```

Submission instructions

After completing this work, submit this word document with your answers to Canvas. In your word document, your answers should be in **bold** print. At the end of your word document, you should write a few sentences that explain the goals of this lab and a paragraph or two that summarizes (high level) the steps one needs to take in order to complete those goals.

The goal of this lab was to explore how AWS Key Management Service (KMS) and Elastic File System (EFS) work together to secure cloud storage. By using the AWS CLI, I created a KMS key and applied it to encrypt an EFS file system, ensuring that data stored in the file system is protected both at rest and during access. This lab demonstrated how to implement encryption using customer-managed keys and highlighted the importance of integrating security best practices into AWS storage solutions.